

Module 3: Data Link Layer and Wireless Networks



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Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

Wireless Networks

- Wireless LANs
- Mobile IP

Hands-On

- Datalink Provider Interface
- SOCK_PACKET
- PF_PACKET

Wireless LANs

Wireless communication is one of the fastest-growing technologies today.

Wireless LANs are widely used in:

- College campuses
- Office buildings
- Airports
- Shopping malls
- Public hotspots



Definition

A Wireless LAN (WLAN) allows devices to communicate without using physical cables.

Introduction to Wireless LANs

Wireless LANs differ significantly from wired LANs.

Major differences include:

- Medium of communication
- Host mobility
- Network architecture
- Access control mechanisms
- Error characteristics

Understanding these differences is essential before studying WLAN protocols.

Wired LAN vs Wireless LAN

Wired LAN	Wireless LAN
Uses cables	Uses air as medium
Point-to-point communication	Broadcast communication
Full-duplex possible	Mostly shared medium
Fixed host connection	Mobile hosts supported
Uses switches	Uses Access Points

Medium Used in LANs

Wired LAN

- Uses copper or fiber cables
- Point-to-point communication
- Full-duplex transmission

Wireless LAN

- Uses air as transmission medium
- Signals are broadcast
- Shared communication channel
- Requires MAC protocols



Wired LAN

- Host connected at a fixed point
- Uses Network Interface Card (NIC)
- MAC address remains unchanged

Wireless LAN

- Host is not physically attached
- Can move freely
- Maintains connectivity while moving

Key Point

Mobility is a fundamental feature of wireless LANs.

Wired Isolated LAN

- Hosts connected through switches
- Communication controlled by switch

Wireless Isolated LAN

- Called an Ad Hoc Network
- Hosts communicate directly
- No switch required

Connection to Other Networks

Wireless LANs are often connected to wired networks.

Examples:

- Internet access
- Campus networks
- Enterprise networks



A special device called an Access Point (AP) provides this connection.

Access Point (AP)

An Access Point acts as a bridge between:

- Wireless environment
- Wired infrastructure network

Functions of AP:

- Receives wireless signals
- Converts wireless frames
- Forwards traffic to wired network
- Manages wireless hosts



Figure 6.1 *Isolated LANs: wired versus wireless*

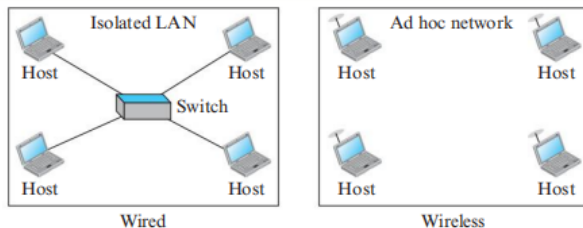
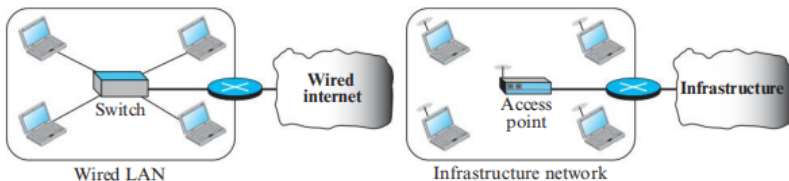


Figure 6.2 *Connection of a wired LAN and a wireless LAN to other networks*



Access Point vs Switch

Switch	Access Point
Connects wired devices	Connects wireless devices
Works only in wired LAN	Connects wired and wireless LANs
Uses cables	Uses radio waves
Layer-2 device	Bridge between two environments

Moving Between Environments

To convert a wired LAN into a wireless LAN:

- 1 Replace wired NICs with wireless NICs
- 2 Replace switch connection with Access Point

Result

- MAC addresses change
- IP addresses remain the same

Important

Only the lower two layers change when moving from wired to wireless communication.

Characteristics of Wireless LANs

Wireless LANs face unique challenges:

- 1 Attenuation
- 2 Interference
- 3 Multipath Propagation
- 4 Errors
- 5 Access Control



Attenuation

Attenuation is the reduction in signal strength as it travels.

Reasons:

- Signal spreads in all directions
- Only a small portion reaches receiver
- Battery-powered devices have limited power

Effect

Weaker signals result in reduced communication range.

Interference

Wireless signals share frequency bands.

Interference occurs when:

- Multiple devices use same frequency
- Signals overlap
- Unwanted transmissions affect reception

Examples:

- Wi-Fi networks
- Bluetooth devices
- Microwave ovens

Multipath Propagation

A signal may reach the receiver through multiple paths.

Causes:

- Reflection from walls
- Reflection from buildings
- Reflection from objects
- Ground reflections

Result:

- Multiple copies arrive at different times
- Signal distortion increases

Error in Wireless Networks

Wireless networks experience more errors than wired networks.

Reasons:

- Attenuation
- Interference
- Multipath propagation

Signal quality is measured using:

$$\text{SNR} = \frac{\text{Signal Power}}{\text{Noise Power}}$$

- High SNR → Reliable communication
- Low SNR → More errors

Access Control

Wireless devices share the same medium.

A mechanism is required to determine:

- Who can transmit?
- When transmission is allowed?
- How collisions are avoided?

This mechanism is called **Medium Access Control (MAC)**.

Why CSMA/CD Does Not Work?

CSMA/CD works well in wired Ethernet.

However, it cannot be used effectively in wireless LANs.

Main reasons:

- 
- 1 No simultaneous transmission and reception
 - 2 Hidden station problem
 - 3 Signal fading over long distances

Reason 1: No Duplex Operation

CSMA/CD requires:

- Sending and receiving simultaneously
- Continuous collision monitoring

Wireless devices:

- Usually battery powered
- Cannot transmit and receive simultaneously

Result

Collision detection becomes impossible.

Reason 2: Hidden Station Problem

Two stations may be unable to hear each other.

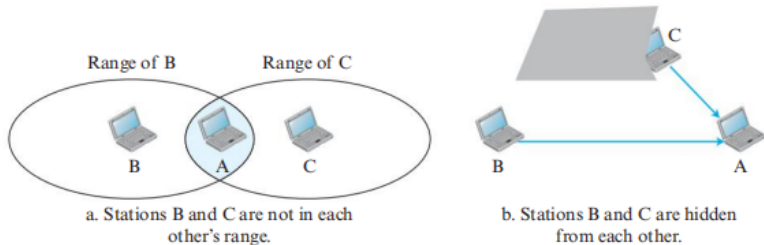
Example:

- Station B communicates with A
- Station C cannot hear B
- C assumes channel is free
- C also transmits to A

Result:

Collision at A

Figure 6.3 *Hidden station problem*



Hidden Station Problem Illustration

B A C

- B is outside C's transmission range.
- C is outside B's transmission range.
- Both can communicate with A.
- Simultaneous transmissions collide at A.

Key Point

B and C are hidden from each other.

Reason 3: Signal Fading

Wireless signals weaken over distance.

Consequences:

- Collision signals may not reach sender
- Collision may occur unnoticed
- Reliable detection becomes difficult

Solution: CSMA/CA

Because CSMA/CD cannot work efficiently in wireless LANs, a new protocol is used.

CSMA/CA

Carrier Sense Multiple Access with Collision Avoidance

Goal:

- Avoid collisions before they occur
- Improve wireless network efficiency
- Handle hidden station problems

CSMA/CA is the standard access method used in modern Wi-Fi networks.

Summary

- Wireless LANs use air as the transmission medium.
- Wireless hosts can move freely.
- Ad hoc networks are isolated wireless LANs.
- Access Points connect WLANs to wired networks.
- Wireless communication suffers from attenuation and interference.
- Multipath propagation causes signal distortion.
- Wireless networks experience more errors than wired LANs.
- CSMA/CD is unsuitable for wireless communication.
- Hidden station problem causes undetected collisions.
- CSMA/CA is used to avoid collisions in WLANs.

Exam Point

CSMA/CD → Wired Ethernet

CSMA/CA → Wireless LANs (Wi-Fi)

IEEE 802.11 Project

IEEE developed the IEEE 802.11 standard for Wireless LANs.

It specifies:

- Physical Layer
- Data Link Layer
- MAC Protocols
- Wireless Communication Architecture



Wi-Fi

Wi-Fi (Wireless Fidelity) is a certification provided by the Wi-Fi Alliance for IEEE 802.11 compliant devices.

IEEE 802.11 Architecture

IEEE 802.11 defines two network architectures:

- 1 Basic Service Set (BSS)
- 2 Extended Service Set (ESS)

Goal:

- Support standalone wireless networks
- Support large interconnected wireless networks

Basic Service Set (BSS)

A Basic Service Set (BSS) is the basic building block of a wireless LAN.

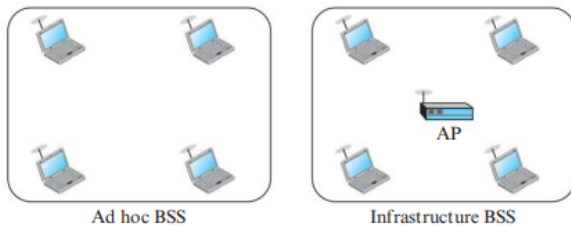
Components:

- Wireless stations
- Optional Access Point (AP)

Two types:

- 1 Ad Hoc BSS (without AP)
- 2 Infrastructure BSS (with AP)

Figure 6.4 *Basic service sets (BSSs)*



Ad Hoc BSS

Characteristics:

- No Access Point required
- Stations communicate directly
- Temporary network formation
- Standalone network

Applications:

- Disaster recovery
- Military communication
- Temporary meetings

Infrastructure BSS

Characteristics:

- Uses an Access Point (AP)
- AP manages communication
- Connection to external networks possible
- Common Wi-Fi architecture

Examples:

- Home Wi-Fi
- Campus Wi-Fi
- Office Wi-Fi

Extended Service Set (ESS)

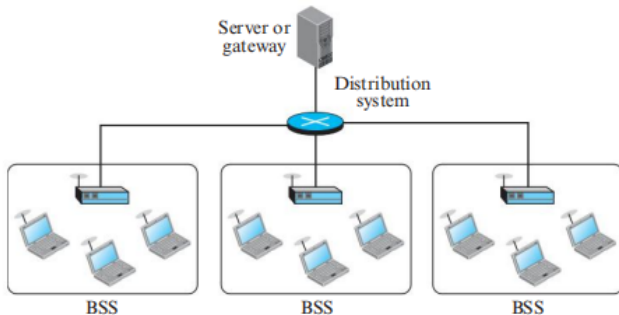
An ESS consists of:

- Two or more BSSs
- Multiple Access Points
- Distribution System (DS)

Distribution System can be:

- Ethernet LAN
- Wireless Backbone
- Other IEEE LAN

Figure 6.5 *Extended service set (ESS)*



Advantages:

- Larger coverage area
- Seamless mobility
- Roaming support
- Better scalability



Example

University Wi-Fi with multiple Access Points connected through Ethernet.

Station Types Based on Mobility

IEEE 802.11 defines three mobility categories:

- 1 No-Transition Mobility
- 2 BSS-Transition Mobility
- 3 ESS-Transition Mobility

Mobility Types

No-Transition Mobility

- Station remains within one BSS
- May be stationary or moving

BSS-Transition Mobility

- Moves between BSSs
- Stays inside same ESS

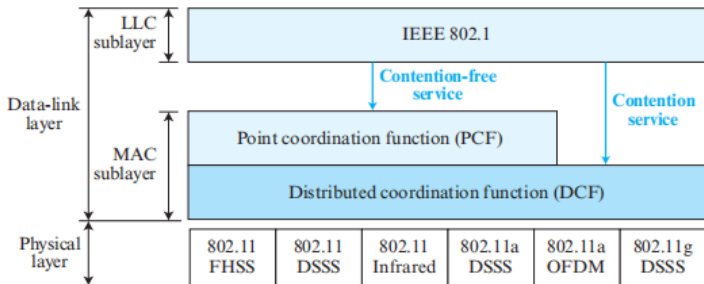
ESS-Transition Mobility

- Moves between different ESSs
- Communication continuity not guaranteed

Two MAC protocols are defined:

- 1 Distributed Coordination Function (DCF)
- 2 Point Coordination Function (PCF)

Figure 6.6 *MAC layers in IEEE 802.11 standard*



Distributed Coordination Function (DCF)

DCF is the mandatory MAC protocol in IEEE 802.11.

Characteristics:

- Distributed control
- Contention-based access
- Uses CSMA/CA
- Supports wireless communication

CSMA/CA = Carrier Sense Multiple Access with Collision Avoidance

Goal:

- Avoid collisions before transmission
- Improve channel utilization
- Reduce retransmissions

Three collision avoidance mechanisms:

- 1 Interframe Space (IFS)
- 2 Contention Window
- 3 Acknowledgment

Interframe Space (IFS)

When a station senses an idle channel:

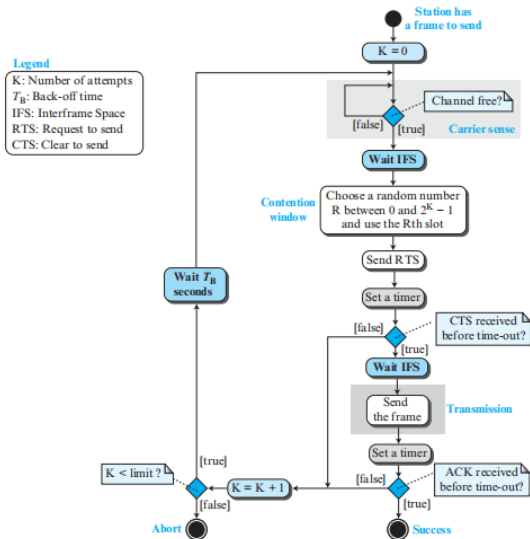
- It does NOT transmit immediately.
- It waits for a fixed period called IFS.

Purpose:

- Allow distant signals to arrive
- Reduce collision probability
- Support priority mechanisms



Figure 6.7 Flow diagram of CSMA/CA



Contention Window

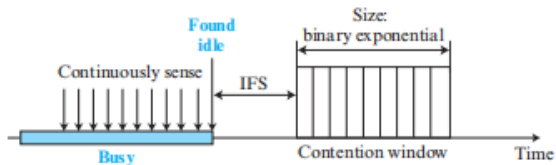
After IFS expires:

- Station selects a random waiting slot.
- Number of slots determined using Binary Exponential Backoff.

Benefits:

- Reduces simultaneous transmissions
- Lowers collision probability
- Improves fairness

Figure 6.8 *Contention window*



Acknowledgment Mechanism

Wireless networks cannot detect collisions directly.

Solution:

- Receiver sends ACK frame
- Sender waits for ACK
- Timeout implies loss or collision
- Retransmission is initiated

Wireless LAN uses handshake frames:

- 1 RTS (Request To Send)
- 2 CTS (Clear To Send)

Purpose:

- Reserve channel
- Reduce collisions
- Solve hidden station problem



Frame Exchange Timeline

Communication sequence:

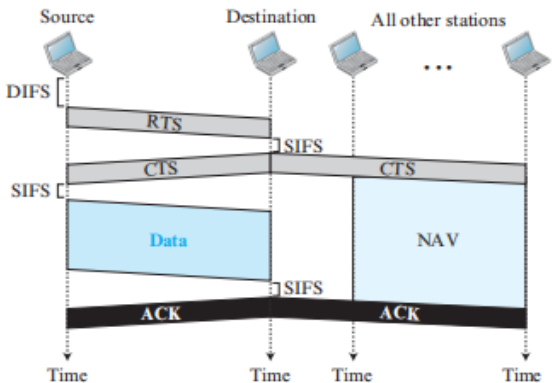
$$RTS \rightarrow CTS \rightarrow DATA \rightarrow ACK$$

Process:

- 1 Source sends RTS
- 2 Destination replies CTS
- 3 Source sends DATA
- 4 Destination sends ACK



Figure 6.9 CSMA/CA and NAV



Interframe Spaces Used

IFS	Purpose
SIFS	ACK, CTS, Response Frames
PIFS	PCF Priority Access
DIFS	Normal DCF Access

Priority:

$$SIFS < PIFS < DIFS$$

Network Allocation Vector (NAV)

NAV is a virtual carrier sensing mechanism.

When RTS is sent:

- Duration field included
- Other stations start NAV timer
- Channel considered busy
- Stations defer transmission

Benefits of NAV

NAV helps:

- Collision avoidance
- Virtual carrier sensing
- Efficient medium sharing
- Reduced hidden-node interference

Hidden Station Problem

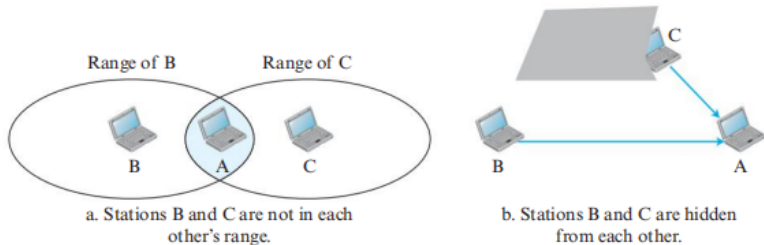
Scenario:

- B sends to A
- C also wants to send to A
- B and C cannot hear each other

Result:

Collision at A

Figure 6.3 *Hidden station problem*



Solution to Hidden Station Problem

RTS/CTS solves hidden station issue.

Process:

- 1 B sends RTS to A
- 2 A sends CTS
- 3 C hears CTS
- 4 C defers transmission



Collision is avoided.

Point Coordination Function (PCF)

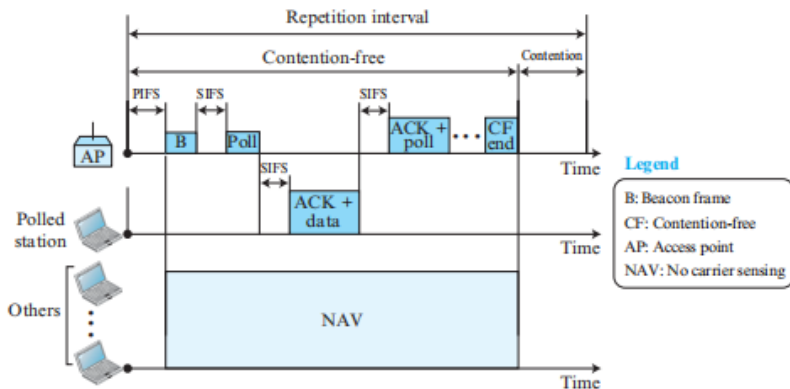
PCF is an optional MAC protocol.

Characteristics:

- Contention-free access
- Centralized control
- Polling-based communication
- Used for real-time traffic



Figure 6.10 *Example of repetition interval*



PCF Operation

Access Point acts as Point Coordinator.

AP performs:

- Polling
- Data reception
- Acknowledgment
- Channel control

No collisions occur during polling.



DCF vs PCF

DCF	PCF
Mandatory	Optional
CSMA/CA	Polling
Contention Based	Contention Free
Distributed	Centralized
Best-effort Traffic	Time-sensitive Traffic

Fragmentation

Wireless channels are noisy.

Large frames:

- Higher error probability
- Expensive retransmission

Solution:

- Divide large frames into smaller fragments
- Retransmit only corrupted fragments

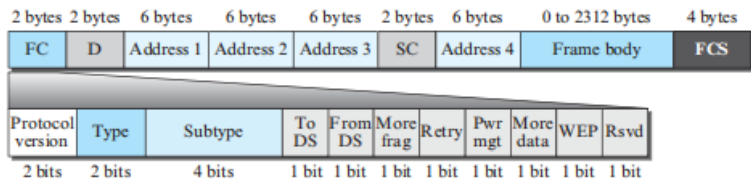
IEEE 802.11 Frame Format

Frame Fields:

- 1 Frame Control (FC)
- 2 Duration (D)
- 3 Address Fields (4)
- 4 Sequence Control
- 5 Frame Body
- 6 FCS



Figure 6.11 *Frame format*



Frame Types

IEEE 802.11 defines three frame categories:

- 1 Management Frames
- 2 Control Frames
- 3 Data Frames



Table 6.1 *Subfields in FC field*

<i>Field</i>	<i>Explanation</i>
Version	Current version is 0
Type	Type of information: management (00), control (01), or data (10)
Subtype	Subtype of each type (see Table 6.2)
To DS	Defined later
From DS	Defined later
More flag	When set to 1, means more fragments
Retry	When set to 1, means retransmitted frame
Pwr mgt	When set to 1, means station is in power management mode
More data	When set to 1, means station has more data to send
WEP	Wired equivalent privacy (encryption implemented)
Rsvd	Reserved

Control Frames

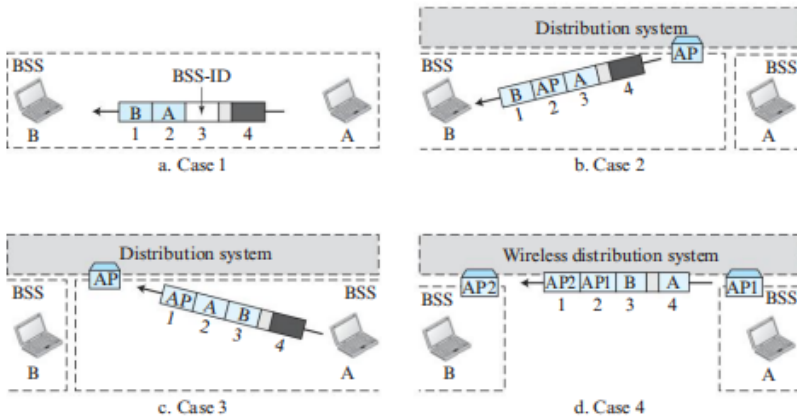
Figure 6.12 *Control frames*



Important Control Frames:

Subtype	Meaning
1011	RTS
1100	CTS
1101	ACK

Figure 6.13 Addressing mechanisms



Exposed Station Problem

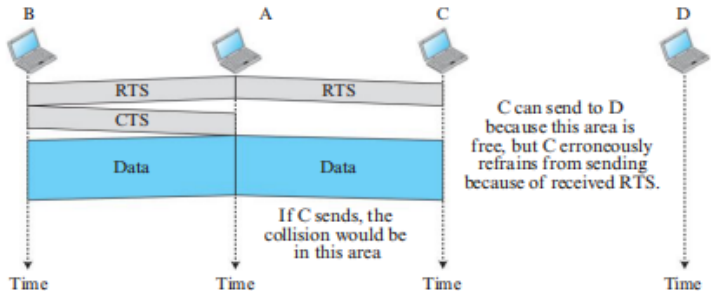
Scenario:

- A sends to B
- C wants to send to D
- C hears A's transmission
- C unnecessarily waits

Result:

- Channel capacity wasted
- Reduced throughput

Figure 6.14 *Exposed station problem*



IEEE 802.11 Physical Layer Standards

IEEE 802.11 defines several physical-layer specifications.

Most implementations operate in the unlicensed ISM bands:

- 902 – 928 MHz
- 2.4 – 2.4835 GHz
- 5.725 – 5.850 GHz



Different standards use different:

- Transmission techniques
- Modulation schemes
- Frequency bands
- Data rates

IEEE 802.11 Physical Layer Specifications

Standard	Technique	Rate
802.11 FHSS	FSK	1–2 Mbps
802.11 DSSS	PSK	1–2 Mbps
Infrared	PPM	1–2 Mbps
802.11a	OFDM	6–54 Mbps
802.11b	HR-DSSS	11 Mbps
802.11g	OFDM	22–54 Mbps
802.11n	MIMO + OFDM	Up to 600 Mbps

FHSS = Frequency Hopping Spread Spectrum

Characteristics:

- Uses 2.4 GHz ISM band
- Band divided into 79 subbands
- Each subband is 1 MHz wide
- Uses pseudorandom hopping sequence

Benefits:

- Reduced interference
- Improved security
- Better reliability

FHSS Modulation and Data Rate

Modulation Techniques:

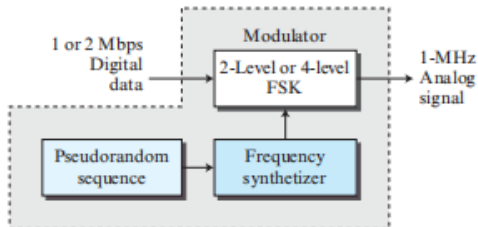
- Two-level FSK
- Four-level FSK

Data Rates:

- 1 Mbps
- 2 Mbps



Figure 6.15 *Physical layer of IEEE 802.11 FHSS*



DSSS = Direct Sequence Spread Spectrum

Characteristics:

- Uses 2.4 GHz ISM band
- Employs spread-spectrum technology
- Resistant to interference
- Better signal integrity

DSSS Modulation

Modulation:

- PSK (Phase Shift Keying)
- Symbol Rate = 1 Mbaud/s

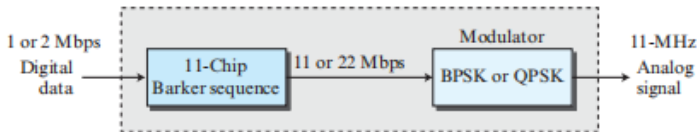
Encoding:

- BPSK $\rightarrow$ 1 bit/ baud
- QPSK $\rightarrow$ 2 bits/ baud

Data Rates:

- 1 Mbps
- 2 Mbps

Figure 6.16 *Physical layer of IEEE 802.11 DSSS*



IEEE 802.11 Infrared

Uses infrared light instead of radio waves.

Operating Range:

$800\text{ nm} - 950\text{ nm}$

Advantages:

- No radio interference
- Secure communication

Disadvantages:

- Requires line-of-sight
- Limited coverage

Infrared Modulation

Modulation Technique:

$$PPM = \text{PulsePositionModulation}$$

For 1 Mbps:

- 4-bit sequence mapped to 16-bit code
- Only one bit is set to 1

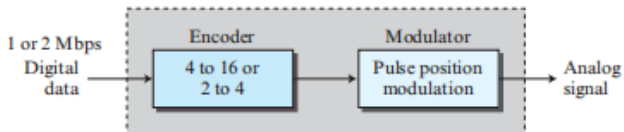
For 2 Mbps:

- 2-bit sequence mapped to 4-bit code
- One bit set to 1

Presence of light = 1

Absence of light = 0

Figure 6.17 *Physical layer of IEEE 802.11 infrared*



IEEE 802.11a OFDM

IEEE 802.11a uses:

OFDM = Orthogonal Frequency Division Multiplexing

Frequency Band:



Advantages:

- Higher throughput
- Reduced interference
- Better channel utilization

802.11a Characteristics

OFDM divides the channel into:

- 52 subbands
- 48 data subbands
- 4 control subbands

Modulation:

- PSK
- QAM

Data Rates:

- 6 Mbps to 54 Mbps
- Common rates: 18 Mbps and 54 Mbps



HR-DSSS = High Rate Direct Sequence Spread Spectrum

Uses:

- 2.4 GHz ISM Band
- Complementary Code Keying (CCK)

Goal:

- Increase throughput
- Maintain compatibility with DSSS

802.11b Data Rates

Supported Data Rates:

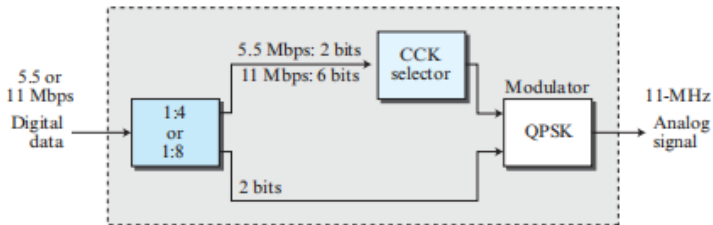
- 1 Mbps
- 2 Mbps
- 5.5 Mbps
- 11 Mbps



Encoding:

- 4-bit CCK for 5.5 Mbps
- 8-bit CCK for 11 Mbps

Figure 6.18 *Physical layer of IEEE 802.11b*



Features:

- Uses OFDM
- Operates at 2.4 GHz
- Forward Error Correction (FEC)
- Backward compatible with 802.11b

Data Rates:

- 22 Mbps
- 54 Mbps

IEEE 802.11n is the next-generation WLAN standard.

Objectives:

- Increase throughput
- Reduce protocol overhead
- Improve reliability

MIMO Technology

MIMO = Multiple Input Multiple Output

Concept:

- Multiple transmitting antennas
- Multiple receiving antennas
- Multiple parallel signal paths

Advantages:

- Improved signal quality
- Better noise immunity
- Increased throughput
- Higher reliability

802.11n Features

Key Features:

- OFDM Based
- Uses MIMO
- Better spectral efficiency
- Enhanced wireless performance

Maximum Throughput:

600 Mbps

Comparison of IEEE 802.11 Standards

Standard	Technique	Band	Rate
802.11 FHSS	FHSS	2.4 GHz	1–2 Mbps
802.11 DSSS	DSSS	2.4 GHz	1–2 Mbps
802.11a	OFDM	5 GHz	54 Mbps
802.11b	HR-DSSS	2.4 GHz	11 Mbps
802.11g	OFDM	2.4 GHz	54 Mbps
802.11n	MIMO+OFDM	5 GHz	600 Mbps

Physical Layer Summary

- FHSS uses frequency hopping.
- DSSS uses spread-spectrum communication.
- Infrared uses PPM and optical signals.
- 802.11a introduced OFDM.
- 802.11b introduced HR-DSSS and CCK.
- 802.11g combines OFDM with 2.4 GHz operation.
- 802.11n uses MIMO for higher throughput.

Exam Focus

FHSS, DSSS, OFDM, HR-DSSS, CCK, MIMO, comparison of 802.11a/b/g/n.